

Fig. 4: Input common-mode voltage range versus supply voltage for AD829

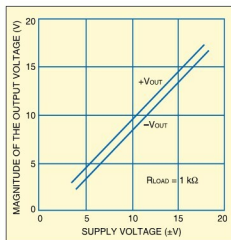


Fig. 5: Output voltage swing versus supply voltage for AD829

The organisation of an op-amp's datasheet is no different from that of discrete semiconductor devices. The information in the datasheet is again presented under the headings of general product description or salient features, absolute maximum ratings, electrical characteristics and thermal characteristics.

performance parameters as a function of one or more relevant parameters. Power dissipation as a function of temperature, CMRR as a function of frequency and closed-loop gain, PSRR as a function of frequency, offsets as a function of temperature and output voltage swing as a function of supply voltage are some examples. In addition, the datasheet includes detailed internal schematic of the device.

Now, let's look at the information contained under the absolute maximum ratings, electrical characteristics and thermal characteristics headings and see what all it has for the op-amp circuit designers.

Absolute maximum ratings

Under the heading of absolute maximum ratings, the datasheet of an op-amp typically includes supply voltage, differential input voltage, input voltage, out-

put short-circuit duration, internal power dissipation, maximum junction temperature, operating temperature range and storage temperature range. It may be mentioned that absolute maximum ratings are stress ratings. Stresses above those listed under these ratings may cause permanent damage

to the device or affect reliability of the device.

Maximum supply voltage is specified for both positive and negative supplies in the case of dual-supply op-amps. Typically, values are in the range of $\pm 18V$ to $\pm 22V$. Operating values are in the range of $\pm 3V$ to $\pm 15V$. Supply voltage parameter is significant as most op-amp parameters depend on it. Available peak-to-peak output voltage swing, input common-mode voltage range, supply current, open-loop gain, slew rate and response time (in the case of comparators) are supply-voltage-dependent. The dependence of these parameters on supply voltage is indicated in the datasheet either in the form of performance curves or by specifying the values of these parameters for a given supply voltage.

Figs 4 and 5 show the dependence of input common-mode voltage range and output voltage swing, respectively, on supply voltage for AD829. In the case of LM156/256/356, the datasheet specifies slew rate of $12 V/\mu s$ and gain-bandwidth of $5 MHz$ for supply voltage of $\pm 15V$ (Table I).

Input voltage parameter quantifies the maximum voltage that can be applied to each of the input terminals. This parameter too is supply-voltage-

TABLE I
AC Electrical Characteristics

$T_A = T_J = 25^\circ C$, $V_S = \pm 15V$

Symbol	Parameter	Conditions	LF155/355	LF156/256/ 356B	LF156/256/ 356/356B	LF257/357	Units
			Typ.	Min.	Typ.	Typ.	
SR	Slew rate	LF155/6 $A_v = 1$, LF357: $A_v = 5$	5	7.5	12	—	$V/\mu s$
			—	—	—	50	$V/\mu s$
GBW	Gain bandwidth product	—	2.5	—	5	20	MHz
t_s	Setting time to 0.01 per cent	(Note 7)	4	—	1.5	1.5	μs
e_n	Equivalent input noise voltage	$R_s = 100\Omega$ $f = 100 Hz$ $f = 1000 Hz$	—	—	—	—	—
			25	—	15	15	$nV/\sqrt{Hz}$
			20	—	12	12	$nV/\sqrt{Hz}$
i_n	Equivalent input current noise	$f = 100 Hz$ $f = 1000 Hz$	0.01	—	0.01	0.01	$pA/\sqrt{Hz}$
			0.01	—	0.01	0.01	$pA/\sqrt{Hz}$
C_{in}	Input capacitance	—	3	—	3	3	pF